

A. M. ISMAIL

Production Engineering Student | Software Enthusiast | Industrial Automation

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🌐 Portfolio 🐙 Github 🔗 LinkedIn

Professional Summary: Production Engineering student specializing in industrial automation and distributed systems. Bridges mechanical engineering fundamentals with software architecture-designing IoT control systems, deploying multi-node compute infrastructure, and building production-grade automation for real industrial environments. Proven track record delivering systems that run 24/7 in manufacturing and academic settings.

Projects

Sri Energy Industrial Automation

[🔗 Project Details](#)

Stack: ESP32, MQTT, FastAPI, React, Pocketbase, NGINX, Docker

- Designed and implemented a three-layer IoT automation system for industrial overhead crane control at a petroleum valve manufacturer, using stateless ESP32 controllers, MQTT messaging, and non-blocking motor logic for emergency stop capability.
- Architected modular MQTT topic schema enabling plug-and-play ESP32 integration without code modification; deployed dual-controller setup with manual/auto failover switch for graceful degradation.

CT Ski ML Cloud Infrastructure

[🔗 Project Details](#)

Stack: Docker Swarm, GlusterFS, JupyterHub, FastAPI, Prometheus, Grafana

- Architected and deployed a multi-user ML compute cluster using Docker Swarm, GlusterFS distributed storage, and custom JupyterHub spawners, serving 16 researchers with 988 compute hours across 247 bookings over 3 months of 24/7 uptime.
- Configured real-time infrastructure monitoring via Prometheus and Grafana stacks; system publicly accessible via Let's Encrypt TLS after upgrading from self-signed certificates. Inaugurated by CEO of Tekion.

SQLC - Quran Competition Management System

[🔗 Project Details](#)

Stack: React, TypeScript, PocketBase, Nginx, Docker Compose

- Engineered a web application suite automating participant registration, dynamic rule-based marksheet templating, and real-time tie-resolution logic.
- Deployed PocketBase with custom JS hooks dynamically injected via a Docker Compose shared carrier volume setup, avoiding heavy image rebuilds.

Middleman - Serverless Sourcing & RFQ Engine

[🔗 Project Details](#)

Stack: React, React Flow, PocketBase, SQLite, Serverless Webhooks

- Engineered an automated RFQ sourcing hub connecting raw email streams to structured database records and an interactive React Flow timeline interface.
- Deployed a lightweight, single-binary PocketBase backend using SQLite, executing data operations and authorization rules with under 50MB of memory.

Virtual Lab for Metrology

[🔗 Project Details](#)

Stack: React, Docker, NGINX

- Designed web-based Profile Projector simulator handling physical measurement operations and laboratory experiments for production engineering students.
- Consolidated 12 decoupled microservices into production ecosystem using NGINX file-based reverse proxy routing.

Additional hobby projects available at codism.in

Internship

SRI Energy Private Limited

June 2026

Industrial Internship | Production & Operations Division

- Completed training in industrial valve assembly and shop-floor inventory operations at the Viralmalai manufacturing facility.

Education

Madras Institute of Technology

Aug 2023 – 2027

IV yr BE in Production Engineering

CGPA: 7.74/10

HSC: 545/600 (2023)

Skills

Software & Tools: MQTT, Docker, Docker Swarm, NGINX, Pocketbase, Prometheus, Grafana, JupyterHub, Git, GlusterFS, Firebase

Programming: Python, FastAPI, React, Flutter, Arduino

CAD & Analysis: SolidWorks, Creo, NX CAD, CATIA, Abaqus CAE

Languages: Tamil, English